MC

EMPLOYER

Maya Chen: projects and experience

IB Diploma · Year 13

Maya Chen

Student builder with hands-on experience in sensors, Python data analysis and running volunteer teams.

SKILLS

- Python (pandas, matplotlib)
- Raspberry Pi and sensors
- Dashboards and data visualisation
- Team leadership
- Public speaking

INTERESTS

Science · Data Science · Computing · Environment · Service

EDUCATION

IB Diploma, Year 13

Grades are not shown in this version.

PROJECTS

Air-quality data report

28 Mar 2026

Cleaned 40,000 readings in Python and found the canteen PM2.5 spike at lunch.

Data Science · Computing · Student-added

Air-quality sensor network

14 Mar 2026

Built 6 Raspberry Pi sensors around campus and a live dashboard.

> *"Maya designed the sensor housing, calibrated every unit against the school weather station and wrote the dashboard herself. Outstanding independent work."*
>
> **Ms Priya Nair, Computer Science**

Computing · Environment · Teacher-endorsed · NIST-1TGBH2

CO_2 sensor prototype

5 Nov 2025

One classroom sensor on a Raspberry Pi, logging every minute.

Computing · Student-added

Eco-committee energy audit

20 Jan 2025

Read every classroom meter for a month; lights left on cost ฿18,000 a term.

> *"Maya translated a vague worry into a costed recommendation."*
>
> **Ms Clara Brandt, CAS Coordinator**

Environment · Teacher-endorsed · NIST-QC76D5

Science fair: microplastics

15 May 2024

Filtered water from three canals and counted microplastic fibres.

Science · Biology · Student-added

Recycling quiz web app

5 Mar 2024

First web app in HTML and JavaScript, used in a Y7 assembly.

Computing · Student-added

Campus composting audit

18 Jan 2024

Weighed canteen food waste for six weeks and proposed a compost bin.

> *“Careful, honest data collection; the bin is still in use.”*
> **Mr David Okafor, Biology**

Environment · Teacher-endorsed · NIST-1QEUO1

LEADERSHIP AND SERVICE

CAS: STEM club for Y7 girls

1 Oct 2025

Co-founded a Friday STEM club; 14 regular members.

> *“A genuinely student-led CAS project with clear impact.”*
> **Ms Clara Brandt, CAS Coordinator**

Service · Teacher-endorsed · NIST-1TQW7D

Mangrove planting trip

15 Jun 2025

Three days planting and tagging mangroves in Chanthaburi.

> *“Maya set up the survival tracking the school now uses.”*
> **Ms Clara Brandt, CAS Coordinator**

Environment · Teacher-endorsed · NIST-1VOTF7

Led the litter survey team

26 Apr 2025

Ran survey days for Beach Guardians and shared the results with the district office.

> *“Her data reached the district office: real-world impact.”*
> **Ms Clara Brandt, CAS Coordinator**

Environment · Service · Teacher-endorsed · NIST-1V4U7T

Tutoring Y8 science

20 Oct 2024

Weekly peer tutoring in the library for two terms.

> *“Reliable every week; her tutees' grades rose.”*
> **Ms Clara Brandt, CAS Coordinator**

Service · Teacher-endorsed · NIST-PS7YZ7

Arduino workshop

5 Oct 2024

Wired a temperature sensor and logged readings to a laptop.

Computing · School-verified · NIST-PI8DA8

Beach cleanup + litter survey

20 Apr 2024

Designed a tally sheet so 40 volunteers could log litter by type.

> *"Maya turned a cleanup into a data-collection exercise the club still uses."*
>
> **Ms Clara Brandt, CAS Coordinator**

Environment · Service · Teacher-endorsed · NIST-1TQPWD

Joined Beach Guardians

20 Sep 2023

Student service club running monthly beach cleanups in Rayong.

Service · School-verified · NIST-1QYTVF

COMPETITIONS AND AWARDS

Youth hackathon: 2nd place

28 Sep 2026

Team of three built a flood-alert chatbot in 24 hours.

Computing · School-verified · NIST-1T6TUH

Data Science summer school

10 Jul 2026

Two-week online summer school; final project on bike-share demand.

Data Science · Student-added

Data visualisation course

10 May 2025

Online course in pandas and matplotlib; 8 graded notebooks.

Computing · Student-added

School-verified **Teacher-endorsed** **Student-added**

Every item shows how it was verified. Students choose and order items but cannot edit verified facts.

NIST International School

Maya Chen · NIST Transit Portfolio

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Each verified item carries a code starting NIST-. Anyone can check a code at **localhost:4961/#/v**. The school confirms what came from its systems or was endorsed by a teacher.